

SDI Intelligence Programme Status

Twenty packs out · the server ordered · Sage X3 starts with estimating

ISSUED 29 SEP 2026

SUPERSEDES 16 SEP 2026

OWNERS J. GRAY · Y. KUMAR · M. YAZIR

COMPILED BY J. GRAY

Status across **SDI Estimating Intelligence**, **SDI Technical Design Intelligence** and **SDI Creative Design Intelligence**, thirteen days on from the third combined report, with the **Sage X3 data migration** and the **SDI Intelligence server** alongside them. The September wave has put **twenty estimate packs** in front of four estimators, and eight detailed reviews have come back. Each one went into the engine as a general rule, not as a correction to one workbook. **The server is being ordered:** a Threadripper PRO build from Scan 3XS, with a three-week lead time and a week's build by James Gray and Jack Calow, so it is in service in the last week of October. It is the blocker named by both other workstreams, and **three services go live on it at the end of October:** AI estimating, Production Design Extraction for the estimators' drawing packs, and Drawing Search. Go-live is confirmed by the parity evidence, not by the date. Yogesh has the **Design Vault core loop working** and a go/no-go set for the end of October. Muhammad's **Client Briefing version 2** is running on Jonathan's live jobs. **Sage X3 now loads and user-tests area by area, starting with estimating** in the week of 5 October.

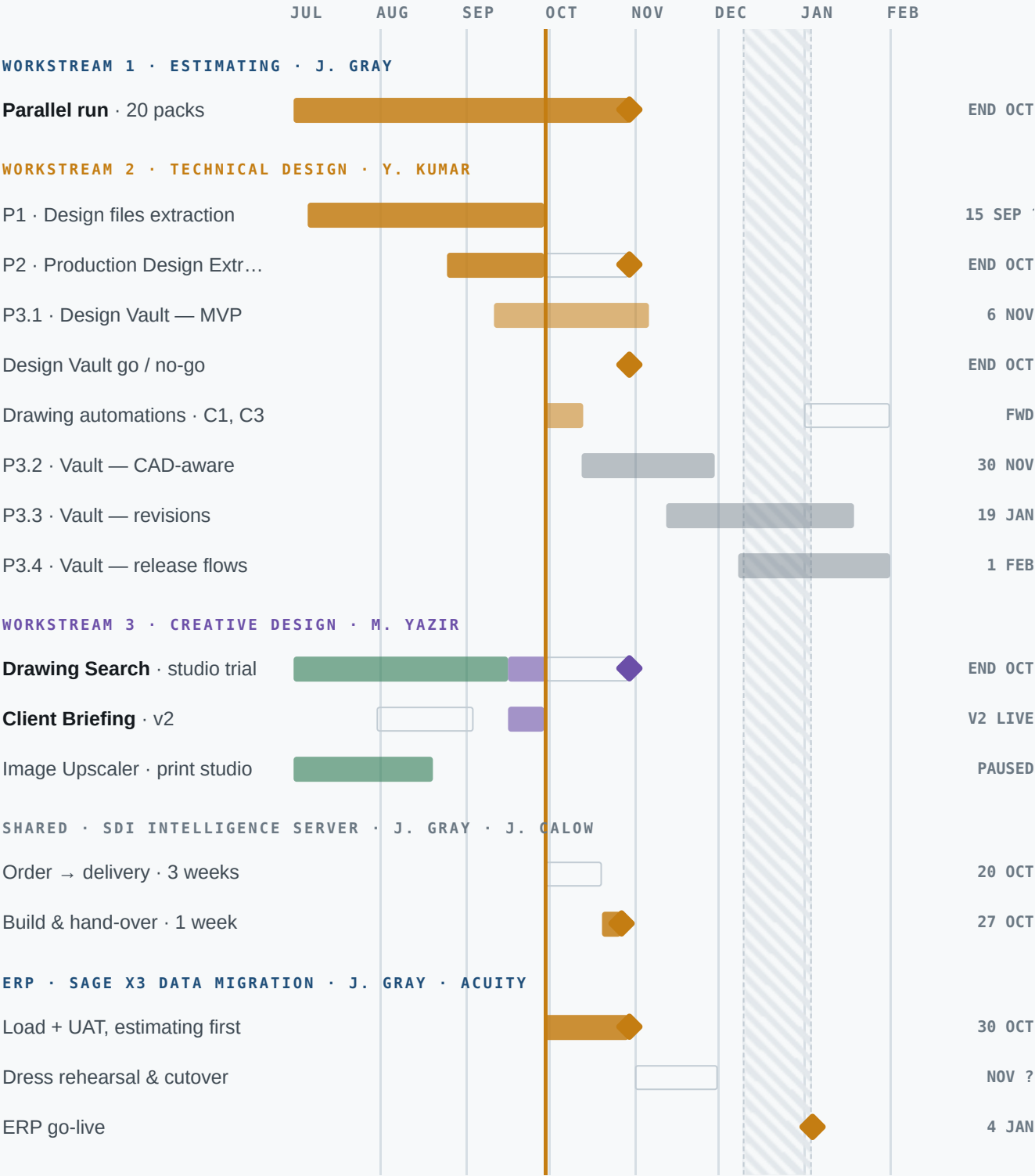
WORKSTREAM	WHAT IT DELIVERS	OWNER	NEXT DATE	STATE
1 · ESTIMATING	SDI Estimating Intelligence Drawings in; costed workbook, BOMs and Routes out	J. Gray	End Oct 26 · with the server	20 OUT · 8 BACK

WORKSTREAM	WHAT IT DELIVERS	OWNER	NEXT DATE	STATE
2 · TECH DESIGN	SDI Technical Design Intelligence Document Manager tools, then the SDI Design Vault	Y. Kumar	End Oct · P2 live · Vault go/no-go	VAULT ~90% UAT WAITS
3 · CREATIVE	SDI Creative Design Intelligence Drawing search on trial · Client Briefing v2 on live jobs	M. Yazir	End Oct · live on the server	ON TRIAL V2 LIVE JOBS
SHARED	SDI Intelligence server Hosts all three · Threadripper PRO 9975WX · Scan 3XS	J. Gray · J. Calow	In service ~w/c 26 Oct	BEING ORDERED
ERP	SDI Sage X3 Data Migration Load and UAT together, area by area	J. Gray · Acuity	w/c 5 Oct · estimating load	TESTED BY 30 OCT

01 The programme on one axis

Three workstreams, the server they share and the Sage X3 migration on one calendar. Re-drawn from Yogesh Kumar's tracker of 25 September, Muhammad Yazir's update, the estimating test log and the Sage X3 e-mails of 25 and 29 September.

ALL WORKSTREAMS · 1 JUL 2026 → 28 FEB 2027



IN PROGRESS DELIVERED DESIGN VAULT CREATIVE DESIGN
WAITING OR TO BE CONFIRMED MILESTONE DEVELOPER LEAVE · 10 DEC — 4 JAN

What moved since 16 September

The server is being ordered. On 16 September it was the one item on every critical path and it had not been ordered. The specification has since been settled and moved up to a Threadripper PRO platform. The reason is that the business has confirmed it will run several things at once on it: one estimate driving SolidWorks, one or two more estimates that do not need SolidWorks, Muhammad's search tool and Yogesh's Design Vault. It is being ordered from Scan 3XS with a **three-week lead time**. **James Gray and Jack Calow**, the admin owners, then need about a **week to build it**. That puts it in service in the week of 26 October. Section 05 has the full specification.

Three go-lives now share one date: the server's. AI estimating, Production Design Extraction (drawing packs into the production area for the estimators) and Drawing Search all go live when the server does, estimated at the **end of October**. That replaces the "September / October" estimating target and P2's 29 September date. In line with Matt's steer of 17 September, **the parity evidence decides go-live, not the date**: the server makes go-live possible, and the AI-against-manual comparisons in section 02 are what confirm it. Two pieces of work are being brought forward in parallel. Yogesh is working on the SolidWorks drawing automations, which were scheduled for January/February, as far as he can. Muhammad and James are analysing how to fast-track the creative design process by integrating automations with the search tool.

The estimating wave doubled. Twenty packs went out in September against eleven at the last report, and eight detailed reviews have come back from four estimators: Howard, Tim, Dave and Tony. The newest jobs are also the largest: a full-bay lit header kit at 163 off, a Tesco external shelter with two roller shutters, and a chiller shelf that estimating logged as three jobs. Dave read the printed 11650-06 book and wrote "*This looks good now*" across it.

Sage X3 changed shape for the better. Loading and user testing now happen in one cycle, one area at a time, on real SDI data. Estimating goes first, in the week of 5 October. All 813 source tables are classified, and every record in 8 of the 13 load templates now carries a status and a reason. Routes are in scope. The dress rehearsal and cutover move to November, to be confirmed.

Technical Design reached a decision point. The Design Vault's core loop (check-in, check-out, locking, versioning) works and was well received by the designers on 23 September. Yogesh has set a **go/no-go at the end of October** on Standard against in-house. The feature that decides it is cross-project and library referencing, and that is what he builds and demonstrates next.

Client Briefing went from two briefs to live jobs. Version 2 was built from the design team's 25 requests, with 19 done and 3 partly done. It now shows Jonathan's current jobs from his inbox, 24 of them today, alongside his calendar.

02

Workstream 1 — SDI Estimating Intelligence

STATUS

Parallel run

OWNER

J. Gray

WITH

James Ryan · Dave's estimators

TARGET GO-LIVE

END OF OCTOBER 2026 · WITH THE SERVER

confirmed by the parity evidence, not by the date

A deterministic cost-estimating engine. A drawing pack goes in, and out come a costed workbook on the estimators' own template, a job report, a covering e-mail, a customer quote and, since 25 September, the **BOMs and Routes** pair. Every priced line names its source. Since the 16 September report there have been 281 commits and 246 numbered decisions (D-083 to D-328), and the suite stands at 7,108 passing.

THE SEPTEMBER WAVE, COUNTED

20 packs issued in September

· one more run and held back

39 packs since May ·

every one on the test log

8 detailed reviews back

· Howard, Tim, Dave, Tony

12 jobs integrated ·

every point now a general rule

246 numbered decisions · D-083

to D-328, each with source and proof

7,108 tests passing

· executing the engine

7 packs awaiting review ·

plus 1 with Dave for comments

0 of 3 supplier price

lists loaded · unchanged

What went out since 16 September

JOB	WHAT IT IS	WHAT IT TAUGHT THE ENGINE	POSITION
401912-02	Tesco metal divider 2 mm mild steel · 1 and 3 off	Drawing, flat DXF and SolidWorks model all read through to the estimate. Dave's reply took 19 minutes: <i>"price would be usable"</i> . His one point was a principle: never price a bought item on a near match, leave it at £0 and flag it. That is now a standing rule.	INTEGRATED · DAVE
M&S concept	Clothes recycling unit budget from a customer render, 350 off	The first costed book from a picture rather than a drawing pack. The rule it taught: an assumption must never look like a measurement. Every sighted size and count is listed for confirmation.	WITH DAVE
11650-06-GA	Boots Coffret hospital kit 2 and 18 off · re-issued with BOMs and Routes	Price only what the drawing number ships, from the kit's own printed parts list. Dave wrote <i>"This looks good now"</i> on the printed book on 28 September, and asked for sub-drawings to come out of material and into labour. That is now a standing layout rule for every job.	INTEGRATED · DAVE
12312-01-GA	Tesco IMS panel-end digital header 1 off, re-run at 200 off	Every line on the drawing is a line on the sheet. Two readings of one table never splice together, and a "P/P" code printed over nine articles gives nine lines.	AWAITING REVIEW
12633-10-GA	Avanti consumable holder clear acrylic · 1 to 200 off	The steel rules carried to acrylic on the first run. The one gap, joining the panels, is now a solvent-weld rule.	AWAITING REVIEW

JOB	WHAT IT IS	WHAT IT TAUGHT THE ENGINE	POSITION
12633-00-GA	Avanti chiller shelf one book for three register lines	Four drawing numbers priced as one product. A plastic assembly is bonded once and the joint is asked; a bend is charged only where the drawing shows it.	AWAITING REVIEW
12614-01-GA	Tesco IMS deep header 132 off	The first pack scored against the drawings before it was run . Four runs in one morning were checked against a written pass table until every row held. Seven rules came out of it, among them that the drawing's callouts outrank a DXF exported short.	AWAITING REVIEW
12567-01-GA	Tesco LED header, full-bay kit 163 off	Bought-in items end to end. A price found is the price charged, and "supplied by others" holds at every stage. Every item the product reaches is charged, free-issued or asked.	AWAITING REVIEW
12645	Tesco DRS external shelter V2 welded steel, two roller shutters · 1 off	The engine was fixed, not the book. The faults found in the 28 September books were fixed in the engine and the pack was re-run, rather than corrected by hand. Eleven general rules came out of it: DWG triage, one drawing is one part, overflow steel counted, a banner that counts items with no line, and department hours read from every labour row. Sent to Dave on 29 September with the shutters and covers named as open.	AWAITING REVIEW

Parity — the go-live KPI

Matt, 17 September: *"The parity data is now the KPI I'm most interested in ... report the variance, key causes and estimator intervention. That evidence should ultimately determine go-live, not the date."* Each job is shown as a variance and a direction. By the pricing policy, no figure from a manual estimate appears in this report.

JOB · ESTIMATOR	AT	VARIANCE	KEY CAUSE	ESTIMATOR INTERVENTION
7332-01 Howard	6 off	-24.8% under	Plating and the freight to the plater carry no price; both are awaiting current quotes, not a figure from a sheet	A 12-point review. Department times and rates adopted throughout
12349-02 Tim	7 off	-22.9% under	Packaging, delivery and two fixings unpriced on that book	14 points; nine confirmed fixed on the revised book
10975-02 Howard	10 off	+12.9% material +0.4%	Material essentially exact; one labour line with Howard to rule on	A line-level review; material now agrees to within a penny
11908-21 Tony	50 off	about +6% over	The joinery route (edging, saw, spindle, bench) was missing and has been added	One review; the rest shown as it is, not tuned to match

JOB · ESTIMATOR	AT	VARIANCE	KEY CAUSE	ESTIMATOR INTERVENTION
1176-02 Howard	10 off	about +33% over	One line: the vinyl graphic, which the customer had already priced at about a quarter of the AI's figure. Without it the AI is about 17% under	One line. The engine had no way to know the customer's price

5 completed comparisons

· of 20 packs this wave

Mean absolute variance across the five is about 20%. On four of the five, the cause is a line with **no price source** (plating, freight, packaging, a customer's own price), not the engine's method. Where the inputs exist, as on 10975-02's material, it agrees to within a penny.

The number grows only

with manual sheets · the ask

A comparison needs the estimator's own sheet for the same job at the same quantity. None has been filed for the twelve packs issued since 17 September. **Ask of Dave's team:** attach the manual sheet wherever one is made, so each week adds comparisons rather than only reviews.

Intervention on the newest books · 12645

Where no manual sheet exists, intervention is measured by what the book leaves for the estimator. The sheet's banner now counts it. On 12645 that is two shutters and the covers to price, 15 provisional steel lines, six market figures, packaging and delivery, one historical match, and two quantity checks.

In the estimators' own words

“Costs come out really close” (Tim, 11350-02); *“price would be usable”* (Dave, 401912-02); *“This looks good now”* (Dave, 11650-06).

The lesson from 1176-02 is structural

A price the customer has already given us sits third in the pricing waterfall, but nothing in a drawing pack tells the engine it exists. It has to be entered somewhere, and the workbook needs a place for it. The same review also handed over two things the engine can adopt: the **powder rate from Access Supply Chain** (POWDER5, Jotun) and a **hanging-geometry method** for powder-coat throughput. Both are the next changes.

What landed in the engine

BOMs and Routes on every run NEW · TIM

Tim asked on 16 September: *"We are looking to Route & Bom this up as well."* Every run now writes a BOMs and Routes pair beside the estimate: a page to read and a workbook to filter. It prices nothing. It answers the two questions that come before any price: what parts are in the pack, and what work each one needs. Each row carries the drawing, page and reader it came from, and quantities are shown both as printed and as rolled up. It is also the natural input to the Sage X3 estimating load in section 06.

A pack is checked against its drawings before it is run NEW · METHOD

From 12614-01 onwards, each pack's drawings are read first and a pass table is written down: the parts list, folds, finishes, what is supplied by others. Each book is then scored against that table. When a book fails a row, the cause is traced and fixed in the engine and the pack is re-run. The book itself is never edited by hand.

Material is confirmed before the run NEW · DAVE

The estimator now confirms material and finish before the engine prices anything. 12645 was the first pack to use this step, with Dave's instruction for standard mild steel.

The sheet tells the truth about what is open NEW · 12645

The banner at the top of the Estimate now counts every open item, not only the priced ones. That includes a bought-in item with no line at all, a stated row the product does not carry, provisional overflow steel and market figures, with the money at stake beside each. Department hours are read from every labour row. The template's own references were corrected, and an audit tool was added for earlier books.

Layout the estimators asked for DONE

The Estimate sheet opens on values, with unused rows folded away (Howard). Parts priced in a material block are no longer repeated as £0 bill lines (Dave). Labour rows name their drawings in their own column, and header values no longer overwrite a label.

Run time is the next piece of engineering NEXT

Large packs such as 12645 take too long end to end. Once this week's jobs are estimated, the work is to time each stage of a run and then re-run only the stages a change affects. The server in section 05 is the other half of the answer.

03 Workstream 2 — SDI
Technical Design Intelligence

STATUS	OWNER	RESOURCE
Vault ~90% proven · P2 UAT waiting	Y. Kumar	Sole developer
DECISION POINT END OF OCTOBER 2026 SDI Design Vault go / no-go · Standard vs in-house		

Two threads. The **Document Manager tools** read SolidWorks project data without a SolidWorks seat and pull the right drawing pack into the estimating app (P1) and the shop-floor Production folder (P2). Then the **SDI Design Vault**, an in-house PDM on GitLab and LFS. Unlike PDM Standard, it has an API that the estimating tools, Sage X3, QC and AI can reach.

The week to 25 September, in Yogesh's report

The Design Vault's core loop works ~90% PROVEN

Browse, lock, edit in a designer WIP folder, check in to GitLab: the full PDM loop runs as a proof of concept, with multi-file check-in, locking edge cases hardened and a **Git-ahead-of-database drift bug** closed. The Vault now has a full observability stack and a new vault explorer (customer, project, file). Its Phase 4 has started: structured CAD references, contains and where-used, and missing-reference display.

The designers' meeting, 23 September (Ed Cooper, Ian B), accepted as working: project creation, auto-versioning, history, check-in and check-out with locking, no-change detection and the WIP/prototype/production traffic light. **The deciding feature is cross-project and library referencing**, which Yogesh builds and demonstrates next.

A firm go/no-go, end of October DECISION

In Yogesh's words, the Vault *"could genuinely succeed — provided the company gives it clear direction and recognises its strategic importance"*. The differentiator over PDM Standard is the API, which lets it link to AI, Sage, QC and the estimating tools where Standard locks the data away. The go/no-go is set for the end of October, subject to discussion with James.

The extraction endpoint, better for estimating IMPROVED

For the 12633 family, the scoping is now correct end to end. A bare run used to return 15 files with silent gaps; it now returns a complete, correctly scoped set for each sub-assembly with no cross-contamination. An assembly-folder token resolves to the assembly it names. A job with no master now fails clearly instead of succeeding hollow. A shared parent drawing's own files are kept out of a sub-assembly run. 582 tests pass, verified live against the W: share. The 11650-06 over-inclusion was traced to a drawing whose Customer Ref held both hands of a mirrored pair.

Asked of estimating: a way to exclude full-unit references when only the kit is quoted, one page per drawing in the PDFs, and confirmation of the exact PDF page estimating needs.

The SolidWorks drawing automations, brought forward

IN PARALLEL

Two automations are proof-of-concept done. **Drawing generation** creates up to about 70% of an SLDDRW drawing from the model (views, sheets, BOM, title block), with the designer still validating the engineering intent and approving it. **Export generation** produces the PDF, DWG, DXF, STEP and IGES release set in full. They were scheduled to start in January/February 2027, after Production Design Extraction and the estimating go-live. **Yogesh now works on them in parallel as far as he can**, to fast-track the design process. No date is claimed while P2's go-live and the Vault decision come first.

PDM Standard, pinned down with Ian

29 SEP

The go/no-go hinges on cross-project referencing, so Yogesh asked Ian B exactly how PDM Standard behaves. The Vault has to match these answers:

- **One local root on every laptop**, the same for everyone, for example `C:\SDI-PDM\Projects\TTi\12689_...`
- **A part owned by another project** (the bracket from released L20, used in R100) is cached to the laptop as a **read-only** copy when the assembly opens, refreshed by Get Latest. Another designer can still check it out in L20.
- **Changing that part from inside R100**: PDM offers to save a renamed copy (if the change is only for R100) or to check it out **from L20**, its owner.
- **If L20 is released**: state change, new revision, make the change, re-release to production.
- **After check-in** others see the new part only after Get Latest (a checkout gets latest automatically). **Both projects still point at the one file in L20**. A released Prototype or Production package does **not** pick up the change; it has to be published again.

Also in Yogesh's week DONE

- **The UI overhauled.** New themes (Steel Blue default), a customer → project → file vault explorer with the customer hierarchy in PostgreSQL, multi-file Add, and pre-submit warnings. GitLab CE is to move to Gitea later to save footprint; the compose file is prepared.
- **Observability:** JSON logs with trace IDs, a health endpoint, a durable event store, Loki/Grafana/Prometheus metrics, and an MCP agent server of 12 read-only tools.
- **Endpoint reach:** it finds the assembly even when the folder name does not match, falls back to a part where no assembly exists, and picks up related DXFs named differently.
- **Requirements captured on 23 September:** an admin layer for absent users' checkouts, library caching, a read-only in-tool view, one-click release to every neutral format, missing-reference validation, a shutdown reminder, get-latest integration with the estimating tools, API hooks, browser viewing for the shop floor, and a migration script from PDM.
- **Documentation:** the PDM Standard vs SDI Design Vault comparison deck, the ADR and requirement register, and the **Technical Design workflow** write-up. The write-up maps creative input → parametric CAD → drawings and PDF → optional client review → export → release → Prototype/Production → QC/ECR, with the Vault as the backbone. It marks what is **ready for UAT** (Prototype/Production extraction), **POC done** (drawing automation, export generation) and **roadmap** (design validation, client review, controlled release and QC/ECR in the Vault).

The plan, as the 25 September tracker states it

REF	PHASE	WHERE IT STANDS	DATE	STATUS
P1	Design files extraction Document Manager API	90% on the tracker. User testing by James from 25 August; deployment still marked not started.	15 Sep 26	NOT CONFIRMED LIVE
P2	Production Design Extraction Document Manager API	95%; development complete and ready for UAT. User testing with Ed and Ian waits on the server ("UAT servers not ready yet"), so it goes live	End Oct 26 was 29 Sep	LIVE WITH SERVER

REF	PHASE	WHERE IT STANDS	DATE	STATUS
		with the server, putting drawing packs in the production area for the estimators. The feature branch merges to main this week.		
P3.1	Design Vault — MVP In-house, phase 1	Development running in parallel with P2; the core loop works as a proof of concept. User testing is planned from 13 October.	6 Nov 26	IN PROGRESS
C1, C3	SolidWorks drawing automations export package, drawing generation	POC done. Were scheduled from January/February; now worked in parallel as far as possible.	Was Jan/Feb 27	BROUGHT FORWARD
P3.2	Design Vault — CAD- aware	Where-used and missing references. Parts of it are already in the Vault's Phase 4 work.	30 Nov 26	NOT STARTED
P3.3	Design Vault — engineering revisions	Approval workflow.	19 Jan 27	NOT STARTED
P3.4	Design Vault — release workflows	WIP, prototype and production release routes.	1 Feb 27	NOT STARTED

Watch-points

- **The server.** Yogesh's report has it "~1–2 weeks, Jack ordering". The order as placed has a three-week lead time and a week's build, so it is in service in the week of 26 October. P2 now goes live with it at the end of October. P3.1's user testing, planned from 13 October, needs its dates re-planned against it.
- **The decision is the company's, not the developer's.** The go/no-go hinges on direction on Standard against in-house, and the Vault needs it to be recognised as a strategic, extensible tool if it goes ahead.
- **One developer, no cover, and more in parallel.** P2's go-live, the Vault's deciding feature and the drawing automations now all run at once through one person. Leave from 10 December to 4 January sits inside the plan.

Next week: merge the extraction branch into main, run a 10-phase review of the repository, and build the cross-project and library referencing feature for the next demonstration.

04

Workstream 3 — SDI Creative Design Intelligence

STATUS

Search on trial · Briefing v2 on live jobs

OWNER

M. Yazir

PROJECTS

2 active · 1 paused

NEXT

STUDIO FEEDBACK · END SEPTEMBER

then changes from it

SERVICE NAME	TRACKER NAME	FOR	POSITION	BLOCKER
SDI Drawing Search Intelligence	Fixture Library	Design and technical teams	TWO-WEEK STUDIO TRIAL	The server – runs on Muhammad's machine
SDI Client Briefing Intelligence	Client Briefing	Jonathan · the studio teams	V2 ON LIVE JOBS	IT for the WhatsApp test
Image Upscaler	Image Upscaler	Sarah, Dean	PAUSED AUG 26	–

Drawing Search: the studio's two-week trial is running TRIAL

The design and technical teams are using it on real work. They give their feedback by the end of the month and Muhammad makes changes from it. It still runs from his own machine, and **the server is its blocker**. The always-on home beside the file server is the one it is waiting for.

Client Briefing: a big week V2 · LIVE JOBS

The design team saw version 1 and liked it. They also said what to change: too many words on screen, and they wanted to see at a glance whether a job is big or small, how much information is in, and whether it is booked. They asked for a view per person, 2D design split from artwork, Jonathan's calendar in the tool, and a way to capture briefs that arrive by phone or WhatsApp. They agreed to start with technical.

Version 2 was built from that feedback and runs alongside version 1 so the two can be compared. Of the **25 things the team asked for, 19 are done and 3 partly done:**

- Every job shows a **size (S/M/L)**, how much information is in, and whether it is booked. Whether a job can start comes from the crucial missing items, not a percentage.
- Each person picks **their own view** (Jonathan, 3D, technical, 2D, artwork, strategy, account manager) and sees only their part.
- Jonathan's **three everyday e-mails are drafted for him** on every job: chasing what is missing, booking the team, confirming it is booked. He still checks and sends them himself.
- Anyone can **ask about a job by typing or by voice**, and the answer is read back.
- **Running on live jobs:** the current jobs from Jonathan's inbox, **24 today**, with new ones picked up automatically at no extra AI cost.
- **Jonathan's calendar**, with his agreement, is in the tool read-only and refreshes every few minutes. It looks like his Outlook and shows who is free or away. He still books in Outlook.
- **Voice and WhatsApp.** After a phone call anyone can speak or upload a voice note. It is transcribed on SDI's own machine, so there is no AI cost and the client is not recorded. The WhatsApp side is built and tested with sample messages. The next test uses a Meta test number, which **needs IT to set up a secure connection**; SDI would then get a proper business number.
- **A new approach underneath.** E-mails are stored once and their key facts extracted once. Planner, writer and checker agents write the brief, and every AI call is logged. The new system has cost **\$2.92 in AI** so far, and the old tool's spend is now logged too.

Nick and Andy have seen it and see a clear opportunity to adapt it for account management. The Client Briefing session with Acuity in section 06 is asked for so that this builds on what X3 already provides rather than duplicating it.

Fast-tracking creative design, with the search tool

IN ANALYSIS

Muhammad and James are working out how to speed up the creative design process by integrating creative-design automations with the search tool, so that a design starts from what SDI has already drawn. It is in analysis; no date is claimed. **Drawing Search itself goes live on the server at the end of October**, with the estimating engine.

Also this period

CARRIED

Muhammad is preparing the documentation and progress for the **LMC meeting in two weeks**, and has registered for an MIT course. The manufacturing business analysis (BOMs and routes into Sage X3, DXF onto CNC) carries on beside the two projects. It now has a direct partner in the Sage X3 estimating load and the manufacturing-actuals week.

Image Upscaler

PAUSED AUG 26 · 56%

Unchanged. Paused on studio priorities, not on any failure; its CMYK conversion stays in daily use.

05

The SDI Intelligence server — ordered

One machine for all three workstreams. It is being ordered this week, with a three-week lead time and a week's build between the two admin owners, James Gray and Jack Calow.

ORDER	PRICE	SUPPLIER
3XS Custom Evolve WRX90E	£13,155.28 inc VAT	Scan 3XS
IN SERVICE		
W/C 26 OCTOBER 2026		
3 weeks' lead · 1 week's build		

COMPONENT	AS ORDERED	WHY
Processor	AMD Ryzen Threadripper PRO 9975WX — 32 cores, 64 threads, 4.0 GHz base, 5.4 GHz boost	Parallel workloads at the same single-core boost SolidWorks and Excel need

COMPONENT	AS ORDERED	WHY
Motherboard	ASUS Pro WS WRX90E-SAGE SE	8 memory channels, 7 × PCIe 5.0 x16 slots, 128 lanes, built-in remote management (BMC)
Memory	128 GB DDR5-5600 ECC registered, 8 × 16 GB (Micron)	All eight channels populated
Graphics	NVIDIA RTX 4000 Ada SFF , 20 GB	SolidWorks-certified; slots and lanes left for a second card
Storage	2 × 2 TB WD Black SN7100 NVMe PCIe 4.0	Two free M.2 slots remain
Cooling	Noctua NH-U14S TR5-SP6 air cooler	Air-cooled for a 24/7 machine
Power	Corsair RM1000e, 1,000 W, Cybenetics Gold	—
Chassis	Fractal Design Define 7, tower	—
Operating system	None in the order; a 16 GB recovery USB is supplied	Licensing is settled separately (below)
Warranty	3 years parts and labour: year 1 onsite, years 2–3 return to base; lifetime 3XS support and upgrade labour	—

WHEN	WHAT HAPPENS	WHO
w/c 28 Sep	Order placed with Scan 3XS	Jack Calow
~w/c 19 Oct	Delivered, after a three-week lead time	Scan 3XS
w/c 19–26 Oct	Built over about a week: operating system and licensing, network and domain, backup, the Linux guests for the Design Vault, SolidWorks, then each workload moved across	James Gray · Jack Calow

WHEN	WHAT HAPPENS	WHO
~w/c 26 Oct	In service, and three go-lives on it: AI estimating (off the laptop), Production Design Extraction for the estimators, and Drawing Search (off Muhammad's machine). Design Vault user testing can start	All three workstreams

Estimating engine · J. Gray

Moves off the laptop. One estimate drives SolidWorks while one or two more run without it, so a long pack no longer holds up the queue.

Drawing Search · M. Yazir

Always-on, beside the file server, instead of on Muhammad's machine. That is the production home the studio trial is waiting for.

Design Vault / PDM · Y. Kumar

GitLab and PostgreSQL guests, and the test server for P2 and P3.1 user testing with the designers.

Why the specification moved from the September request

The 3 September request was a single Ryzen 9 9950X box at around £5,000 ex VAT, sized for the fastest single core. Since then the business has said it will run several things at once on this machine: an estimate driving SolidWorks, one or two more that do not need it, the search tool and the Vault. A Threadripper PRO keeps the same single-core boost for SolidWorks. It adds four times the memory bandwidth and PCIe lanes, and a socket that takes more cores later without replacing the machine. The order is about £10,963 ex VAT.

To settle during the build week

- **Operating system and licensing.** None is in the order. Windows Server licensing at 32 cores needs a second core pack, and SolidWorks needs a supported Windows desktop. Which runs on the metal, and which as a guest, is decided before the build.
- **Disks mirrored** (RAID 1) and a backup regime, since this becomes the business's machine.
- **A UPS**, and placement on the same LAN as the file server and sdi-dc01.
- **A second GPU** if the Design Vault or local models need one. The board has the slots and lanes.

06 Sage X3 data migration

STATUS

Load + UAT, estimating first

OWNER

J. Gray

WITH

Acuity · Dave's team · the data owners

ALL AREAS LOADED AND USER-TESTED

FRIDAY 30 OCTOBER 2026

dress rehearsal and cutover in November, to be confirmed

The Access Supply Chain database behind SDI Live moves to **Sage X3**. Since 16 September the approach has changed. **Data loading and user acceptance testing are one cycle**, area by area: each area is loaded into test, tested by its own users on real SDI data, fixed, and then kept current by incremental updates. That means UAT runs on the data SDI will actually go live with, and Acuity's standard reports list becomes the test scripts.

Estimating goes first

The first full pass is **products, prices, BOMs and routes**, loaded and user-tested end to end in X3 with Dave's team. It covers:

- **Functions today against X3.** How Dave's team creates prices and imports BOMs and routes in Access, and the X3 equivalents. Dave's template is the starting point, and the aim is to improve on it, not copy it.
- **Pre-populated tables.** The X3 tables that must exist before an estimate can be uploaded: product categories, units, work centres, labour and cost rates, price structures. Acuity has been asked for the list, with templates for any SDI supplies.
- **Loading SDI's data** into test: current products, work centres, BOMs (lowest level first), routes and prices.
- **UAT on estimates.** Sample estimates uploaded with their BOMs, routes and prices, viewed, and run through the standard reports. This includes how multi-level BOMs and routes are created and maintained in X3, and **how the estimating tools connect**.
- **Manufacturing actuals.** Actual routes and times from production loaded back, so that estimated and actual can be compared.
- **Incremental updates.** Only the changes from Access are loaded, not a full dump and reload, which keeps test current through to cutover.

Where this meets SDI Intelligence. The engine's BOMs and Routes output (section 02) is an estimate's BOM and route in a structured form, which is the shape an X3 upload takes. How the estimating tools connect is now an explicit UAT item, rather than something to solve after go-live.

All 813 source tables, classified

CLASS	TABLES	WHAT HAPPENS
Carry the migration	21	Loaded to X3, in seven areas: products and stock, BOMs and routes, works orders, sales, purchasing, GL, AR/AP
Need a business decision	16	10 decisions in all, each with an SDI owner
Custom-field (UDEF) tables	18	Scoped from the reports users need
History	41	Kept in the SDILive archive

CLASS	TABLES	WHAT HAPPENS
Duplicates	40	Including dated backup copies; dropped
Empty, system, transient or rebuilt in X3	677	MRP output, print and posting work tables and the like. Whole areas such as MRP planning, CRM and Access's own reporting drop out because X3 rebuilds them

This replaces the "131 need a business decision" of the first 25 September e-mail. The sixteen decision tables resolve into ten decisions. Five of those depend on X3 configuration: cost centres (the priority, because they decide how the GL opening balances load), multiple delivery addresses per customer, analysis codes, pricing and operators.

Data quality — every record with a status and a reason

Done for 8 of the 13 load templates: products, BOMs, routes, works orders, customers, suppliers, open AR and open AP. Stock, open sales and purchase orders, product sites and GL balances follow once their rules are agreed.

AREA	IN ACCESS	WHAT THE ANALYSIS FOUND
Products	41,645	About 16,400 current ; 7,399 active two to five years ago need a keep-or-drop decision; 17,841 have no activity
Descriptions	10,738	Current products with no main description. 10,687 have the text in another description field , which is fixed in the mapping. Only 51 need a description added
BOMs	30,745 current	13,116 lines to review, 20,099 with no activity
Routes	31,232 current	Operations. Only 68 are missing times
Works orders	936 open	918 have no end date , which Acuity confirms is mandatory; one rule from Manufacturing clears them. 12 need a route
Customers	217 good	Active and complete. 52 possible duplicates, 28 incomplete, 202 not traded in two years

AREA	IN ACCESS	WHAT THE ANALYSIS FOUND
Suppliers	623 good	Active and complete. 130 incomplete (mostly addresses or postcodes), 492 not used in two years
Open AR	730 items £8.51m net	Every item belongs to a customer that is loading; nothing is over 12 months old
Open AP	2,014 items £4.97m net	The same. The only gap is due dates on payments on account, which default to the document date

What loads, and what stays behind

SDI expects to load roughly **40% of the product, BOM and route data** and around **60% of customers and suppliers**. The rest stays in the SDILive archive, which is taken in full, verified and queryable before Access is switched off.

Each record sits in one **bucket**, and each bucket has its own route to X3. **Good** records load straight away. **Incomplete** records are fixed in the mapping or by the owner in Access. **Possible duplicates** are merged or confirmed by the owner. **Review** records are kept or dropped by the owner from a list. **Old** records are dropped as a confirmed block. The good bucket of each area loads first and is tested while the owners work the other buckets, so test gets bigger and cleaner each week.

Acuity's new templates, mapped field by field

From `ASC_SageX3_NewTemplate_Mapping.xlsx`. Every field of Acuity's new YA004 load templates is mapped to its Access source, with the rule, any fixed value, and whether it is waiting on Acuity.

121 fields mapped ·
source column and rule

47 fixed values · site
S01A, ledger 18L, doc types

17 to review · rule to agree

22 blocked on Acuity
· configuration values

TEMPLATES	POSITION
ITF, TXT1–4	Product sites and the four product texts: ready once ITM loads
ITM	Products: needs the Access product group translated to X3 category codes (TCLCOD), built with Acuity
BOM	Manufacturing BOMs: orphan check first
BPC, BPS	Customers and suppliers: waiting on Acuity's tax-rule, accounting, category and payment-terms codes
SOH, POH	Open sales and purchase orders: views built. The new template changes the order type (SON), numbers lines in thousands and dates as YYYYMMDD. Nine view changes, marked to fix now
BICX, BISX, GAS	Open AR, open AP and the GL trial balance: the new templates are a new structure , so the three views are rebuilt from the mapping rather than patched
SMR	Stock opening balances: waiting on a stock-flag fix; four view changes (transaction type 19, date format, unit price, location)
BUD	Budgets: new in scope for review; budget code, version and fiscal year to confirm

Thirteen questions back to Acuity, four of them critical

Critical, because nothing loads without them: the X3 site code (S01A, to confirm for this implementation), the company code, the product category codes with their mapping from Access, and the unit-of-measure codes. **High**: tax rules, accounting codes, customer and supplier categories, payment terms, and the AR and AP control accounts. **Medium**: the three GL dimensions, the budget code and fiscal year, and the stock opening posting date.

Control totals agreed before load, checked after it. Open sales orders: 1,159 lines on 342 orders, £13,535,254. Open purchase orders: 1,166 lines on 338 orders, £5,675,222. AR and AP are re-read at load (29 September: 730 items, £8.51m net, and 2,014 items, £4.97m net). Stock, the trial balance, items, BOMs, customers and suppliers are counted once their views are final. Each total has an owner who signs it.

Routes are not among the sixteen. The route template is still on the list asked of Acuity, and the estimating pass in the week of 5 October needs it.

The plan to 30 October

Provisional and for agreement with Acuity. The estimating kick-off is not waiting on it.

WEEK	WHAT HAPPENS
w/c 28 Sep	Kick-off call. Agree the tables to pre-populate. Test environment access. Mapping fixes. Key decisions: works-order end dates and the activity cutoff
w/c 5 Oct	Load the estimating data ; UAT on estimates starts
w/c 12 Oct	Estimating signed off. First incremental update. Load customers and suppliers; UAT starts on them
w/c 19 Oct	Manufacturing actuals. Load stock, open orders, open AR and AP, GL balances, with UAT
w/c 26 Oct	Works orders. All areas loaded and user-tested by Friday 30 October
November	Dress rehearsal and cutover, dates to be confirmed
4 Jan 27	ERP go-live. From it the estimating engine reads material and supplier prices from X3, with dual-running across the changeover

Resolved with Acuity CLOSED

- Open items are sent positive, with the INV/CRN/PAY type driving the balance, so AR and AP are unblocked
- One site, S01A, for finance and stock
- The chart of accounts is already in X3
- Product categories: FGSL, FGML, PHANT, RAWML
- BOMs follow products; multi-level loaded lowest level first; the 32 orphan parents excluded
- Obsolete products, templates and placeholders dropped
- **Routings are in scope**, which closes the 16 September question

Still needed from Acuity ASKED

- The list of tables to pre-populate before an estimate uploads, with templates
- Import templates for routes, open AR (BICX), open AP (BISX) and the GL trial balance (GAS), and confirmation of the stock and open-order templates
- Whether work centres are a separate master load before routes
- Mandatory fields: are blank descriptions or zero prices rejected?
- Test environment access and refresh; the recommended cutover method
- Their recommended reporting approach (SEI or native), to map the 18 custom-field tables

Reporting, and two sessions asked for

Acuity's standard reports list is mapped against what users need and used in each area's UAT, alongside an inventory of SDI's Crystal Reports. Tracing each required report back to its source data decides which of the 18 custom-field tables matter.

BOMs and routes in X3, with Dave and his team: how they are structured and maintained, how multi-level BOMs behave, and how a quoted job becomes a manufacturing BOM and route. This is the hand-over from estimating to manufacturing. **Client Briefing**, with Andy Baxter, Stuart McMurray and Muhammad, so that the AI work on briefing builds on what X3 already provides.

07 Open across the programme

Ordered by what it costs to leave alone. Items closed since 16 September are listed last.

For Matt: the critical path, resource, and the decisions that are his

Matt asked on 17 September to be told specifically about anything that threatens the critical path, needs more resource or needs his decision.

- **Decision: the server's cost.** The workstation approved in September was specified at about £5,000 ex VAT. The order is **about £10,963 ex VAT (£13,155.28 inc)**, because it now has to run the three services and the Vault side by side rather than one engine. It is being ordered on the existing approval. If the difference needs Matt's sign-off, that is the one decision in front of the order.
- **Critical path: the server's delivery.** Three services go live on it at the end of October. A slip in the three-week lead time or the build week moves all three together.
- **Decision: the Design Vault, end of October.** Standard against in-house is a company direction, not a technical call. Yogesh's evidence will be the cross-project referencing demo measured against Ian's answers above.
- **Resource: one developer carrying three things.** P2's go-live, the Vault's deciding feature and the drawing automations now run in parallel through Yogesh, with December leave inside the plan. More hands are the only way to hold all three dates.
- **Resource: Dave's team in October.** The same estimators are asked for the seven outstanding reviews, the manual sheets that grow the parity KPI, and the Sage X3 estimating UAT from the week of 5 October. That is the programme's most contended resource. It needs protected time, or an agreed order of priority.

The server: order placed, then a week's build ACT

Place the Scan 3XS order this week. Before it arrives, settle the operating system and licensing, RAID and backup, a UPS and the network placement, so that the build week is spent building. Every workstream's next step waits on it: P2 and P3.1 user testing, Drawing Search's production home, and estimates leaving the laptop.

All three workstreams. Owners: James Gray and Jack Calow.

Sage X3: this week's decisions ACT

The works-order end-date rule from Manufacturing (918 orders), the activity cutoff for products and BOMs, the tables to pre-populate, and test environment access. From Acuity, the four critical configuration answers (site, company, category and unit-of-measure codes) and the route template. The estimating load in the week of 5 October depends on all of them.

Sage X3. James Gray with Acuity; the end-date rule from Manufacturing.

Reviews back on the seven awaiting ACT

12552, 12312-01, 12633-10, 12633-00, 12614-01, 12567-01 and 12645 are with Tim, Howard and Dave, and the M&S concept is with Dave for comments. Every returned review has become rules, so the go-live date moves on these.

Workstream 1. Ask of Dave's team.

Grow the parity comparisons every week ACT

Five completed AI-against-manual comparisons so far. Each new one needs the estimator's own sheet for the same job at the same quantity. This is the evidence that confirms the end-of-October go-live.

Workstream 1. Ask of Dave's team: the manual sheet with every job that has one.

P3.1's user-testing dates, against the server RE-PLAN

P2 now goes live with the server at the end of October. P3.1's user testing, planned from 13 October, needs the same server and is to be re-planned. P1 is still not confirmed live, with its deployment step marked not started.

Workstream 2. Yogesh Kumar.

Design Vault go/no-go, end of October DECISION

Standard against in-house, decided on the cross-project and library referencing demo and on the company's direction.

Workstream 2. James Gray with Yogesh Kumar, Ed Cooper and Ian B.

A secure connection for the WhatsApp test IT

Client Briefing's WhatsApp side is built and tested with sample messages. The Meta test number needs IT to set up a secure connection; then SDI needs a proper business number.

Workstream 3. IT with Muhammad Yazir.

A place in the workbook for a customer's own price OPEN

1176-02 showed that one line the customer had already priced outweighed the whole variance. Also from that review and ready to adopt: the Jotun powder rate from Access Supply Chain, and bar-occupancy for powder-coat throughput.

Workstream 1.

Supplier price lists are still not loaded OPEN

The ingester is built and tested; it needs the actual files from Elite, Eagle and Thermaset. Every catalogue loaded replaces market figures with reproducible prices, and every book in this wave carried some.

Workstream 1. Carried from 27 August, unchanged.

Run time on large packs NEXT

Time each stage of a run, then re-run only what a change affects. The work starts once this week's jobs are estimated.

Workstream 1.

Carried, with no change reported CARRIED

- **The BrightHR client secret has never been rotated.**
- **SDI-APP01** still needs the rotated database password and its build configuration.
- **Two price sources wired to empty tables** (rung 3 of the price chain).
- **Three configured rates with no live source** (board sheet, zinc plating, section stock), awaiting a ruling.
- **The W: drive and the Vault:** syncing W: on released status is a design decision for P3.1.
- **Which machines get a SolidWorks seat**, now including the server.

Closed since 16 September

CLOSED

- **The workstation: quotes finalised, decisions taken, order placed.** The quotes are finalised and the specification settled; the order is going in this week (section 05).
- **Routings and works orders: in scope or parked?** In scope. Routes are one of the eight templates with record-level results.
- **Sage X3's three-week "mobilise, cleanse, cut over" plan** is superseded by load and UAT by area, with cutover in November.
- **20,724 blank item descriptions** is superseded. Of the 10,738 current products with no main description, all but 51 have the text in another field.
- **Jonathan's time on real briefs:** Client Briefing v2 runs on his live jobs and his calendar.
- **The estimating go-live target:** restated as the end of October, with the server, and confirmed by the parity evidence.
- **Packaging and delivery held at £0:** they are now priced as market figures and named for the estimator to confirm (£135 and £95 on 12645).

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SOURCES · SDI INTELLIGENCE COMMIT HISTORY 16 – 29 SEP AND CHANGE REGISTER D-083 TO D-328 · SDI INTELLIGENCE ESTIMATING TEST LOG AND TESTING STATUS, 29 SEP · ESTIMATOR REVIEWS FROM HOWARD THURLEY, TIM WILKES, DAVE WRIGHT AND TONY FORD · YOGESH KUMAR STATUS AND PROJECT PLAN TRACKER 25 SEP, PDM STANDARD CLARIFICATION WITH IAN BEAUMONT 29 SEP, TECHNICAL DESIGN WORKFLOW WRITE-UP · MATT EVANS 17 SEP · MUHAMMAD YAZIR WEEKLY UPDATE · SCAN 3XS CONFIGURATION 29 SEP · SAGE X3 E-MAILS TO ACUITY 25 AND 29 SEP

ESTIMATING GO-LIVE TARGET · END OF OCTOBER 2026, WITH THE SERVER – CONFIRMED BY THE PARITY EVIDENCE, NOT BY THE DATE

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